

Unit 15: Earth's Dynamic Surface — Volcanoes

Reading: pages 212–225 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. At convergent boundaries, magma is formed primarily by
 - a) melting from friction between the plates
 - b) heat from radioactive decay
 - c) decompression melting
 - d) flux melting
2. Felsic magmas tend to erupt explosively mainly because they are
 - a) much hotter than mafic magmas
 - b) more viscous and richer in volatiles, so gases cannot escape easily
 - c) denser than mafic magmas
 - d) lower in silica than mafic magmas
3. Which type of volcano is almost always associated with subduction zones?
 - a) Shield volcano
 - b) Cinder cone
 - c) Composite volcano
 - d) Large igneous province
4. Shield volcanoes have very gentle slopes, only about 2° to 10°, because
 - a) their mafic magma has low viscosity and flows a long way before cooling
 - b) they erupt only rarely
 - c) they are built from loose fragments that cannot pile steeply
 - d) they form underwater where slopes cannot steepen
5. A *lahar* is
 - a) a fast-moving cloud of hot gas and tephra
 - b) a mudflow or debris flow related to a volcano
 - c) a ropy texture formed by slow-moving lava
 - d) a volcanic crater more than 2 km in diameter

6. A volcanic crater more than 2 km in diameter is called
- a) a caldera
 - b) a vent
 - c) a lava tube
 - d) a cinder cone

Part B · True or False

2 questions × 1 point = 2 points

7. Historically, indirect volcanic hazards have killed far more people than direct hazards.
TRUE / FALSE
8. Pahoehoe forms when lava is forced to flow faster than it easily can, producing a blocky, jagged surface.
TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. Compare cinder cones, composite volcanoes, and shield volcanoes. For each, give its approximate size, its usual magma composition, and its typical eruption style.
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10. Explain how the Hawaiian island chain formed, and what such a chain reveals to geologists about plate motion.
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